

Amber Horvath

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Education

Carnegie Mellon University – Ph.D. in Human-Computer Interaction, 2024.
Thesis Title: Meta-Information to Support Sensemaking by Developers
Advisor: Brad A. Myers
Thesis Committee: Laura Dabbish, Elena Glassman, Aniket Kittur, and Andrew Macvean

Carnegie Mellon University - M.Sc. in Human-Computer Interaction, 2021.

Oregon State University - B.Sc. in Computer Science, 2018.

Professional Experience

ASSISTANT PROFESSOR, UNIVERSITY OF VICTORIA – 2026-PRESENT

- Co-PI of the CHISEL Group. Deployed and evaluated CLAI in a qualitative expert design review [I1]. CLAI is an agentic code comprehension architecture to probe on how comprehension work should be allocated between developer and LLM agent during LLM-mediated software development.

POST-DOCTORAL ASSOCIATE, MASSACHUSETTS INSTITUTE OF TECHNOLOGY – 2024-2026

- Performed a qualitative diary study to codify signals that a developer is blocked by a lack of knowledge about code and what steps are taken to resolve these information blockages.

- Used results of diary study to inform the design of CLAI [I1] and inspire the design probe for supporting onboarding as a first-class comprehension task.

PHD RESEARCHER, HCII, CARNEGIE MELLON UNIVERSITY – 2018-2024

- Built research prototypes using developer interaction traces as a form of meta-information to serve as comprehension aides when programming by: supporting documentation usage [C2, C4], code navigation [C1, C3], and debugging with AI support [I2].

- Investigated API usability challenges including discoverability of functionality [C5], nascent mental models of the problem space [E1], and dealing with boilerplate code [C7].

- Assisted on qualitative analysis for understanding data scientists' challenges when finding past analyses choices [C6].

- Collaborated with researchers at Google to integrate industry insights and considerations into research designs and prototypes, resulting in six co-authored publications including one award-winning paper [J1].
- Mentored 8 undergraduate and graduate students in techniques for developing high-quality code for easier collaboration and completing qualitative research, resulting in six students appearing as co-authors on two published papers.

RESEARCH INTERN, GOOGLE – 2019

- [J1] Designed and completed a qualitative inquiry using card sorting of developer's sentiment and perceived "promotability" towards common low-level software engineering tasks at Google (e.g., writing documentation) to determine what tasks are seen as non-promotable.
- Completed a regression model in R of how often the "most" non-promotable task, code review, is completed by engineers of diverse genders.

USER-EXPERIENCE INTERN, INTEL – 2017

- Designed user interfaces for mobile applications and assessed usability through qualitative user studies (interviews, think-aloud studies).

UNDERGRADUATE RESEARCH INTERN, OREGON STATE UNIVERSITY, CARNEGIE MELLON UNIVERSITY – 2013-2017

- Completed design inquiry, development and prototyping of a tool to make data science programming more user-effective through lightweight code versioning [C9].
- Assisted in the development and evaluation of tools to assist novice developers learn core programming concept and professional developers navigate to find bugs [C12] .
- Assisted in the application and deployment of GenderMag to understand diversity inequities in open-source and professional software development [J2, C8, C10].

USER-EXPERIENCE INTERN, HEWLETT-PACKARD INC. - 2015

- Designed prototype interfaces for mobile printing.

Publications

IN-SUBMISSION

[I1] ***CLAI: Programmable Support for Code Comprehension in LLM-Assisted Development.*** Amber Horvath and Arvind Satyanarayan. Submitted to CHI 2027.

[I2] ***Lazuli: Material Interaction Support for Print Debugging with a Multi-Modal LLM Interface.*** Amber Horvath, Andrew Macvean, and Brad A. Myers.

JOURNAL ARTICLES

[J1] ***Systemic Gender Inequities in Who Reviews Code.*** Emerson Murphy-Hill, Jillian Dicker, Amber Horvath, Maggie Morrow Hodges, Carolyn D. Egelman, Laurie R. Weingart, Ciera Jaspan, Collin Green, and Nina Che*n,* **CSCW '23.** [26% acceptance rate, Diversity and Inclusion Award Winner]

[J2] ***How Gender-Biased Tools Shape Newcomer Experiences in OSS Projects.*** Hema Susmita Padala, Christopher Mendez, Felipe Fronchetti, Igor Steinmacher, Zoe Steine-Hanson, Claudia Hilderbrand, Amber Horvath, Charles Hill, Logan Simpson, Margaret Burnett, Marco Gerosa, and Anita Sarma, **TSE '20.** [12% acceptance rate]

[J3] ***General Principles for a Generalized Idea Garden.*** William Jernigan, Amber Horvath, Michael Lee, Margaret Burnett, Taylor Cuilty, Sandeep Kuttal, Anicia Peters, Irwin Kwan, Faezeh Bahmani, Amy Ko, Christopher J. Mendez, and Alannah Oleson, **JVLC '17.**

CONFERENCE PROCEEDINGS

[C1] ***Meta-Manager: a Tool for Collecting and Exploring Meta Information about Code.*** Amber Horvath, Andrew Macvean, and Brad A. Myers, **CHI '24.** [26.4% acceptance rate]

[C2] ***Support for Long-Form Documentation Authoring and Maintenance.*** Amber Horvath, Andrew Macvean, and Brad A. Myers, **VL/HCC '23.** [30% acceptance rate]

[C3] ***Using Annotations for Sensemaking About Code.*** Amber Horvath, Brad A. Myers, Andrew Macvean, and Imtiaz Rahman, **UIST '22.** [26.3% acceptance rate]

[C4] ***Understanding How Programmers Can Use Annotations on Documentation.*** Amber Horvath, Michael Xieyang Liu, River Hendriksen, Connor Shannon, Emma Paterson, Kazi Jawad, Andrew Macvean, and Brad A. Myers, **CHI '22.** [26% acceptance rate]

[C5] ***The Long Tail: Understanding the Discoverability of API Functionality.*** Amber Horvath, Sachin Grover, Sihan Dong, Emily Zhou, Finn Voichick, Mary Beth Kery, Shwetha Shinju, Daye Nam, Mariann Nagy, and Brad Myers, **VL/HCC '19.** [30% acceptance rate]

[C6] ***Towards Effective Foraging by Data Scientists to Find Past Analysis Choices.*** Mary Beth Kery, Bonnie E. John, Patrick O'Flaherty, Amber Horvath, and Brad A. Myers, **CHI '19.** [23.8% acceptance rate]

[C7] ***MARBLE: Mining for Boilerplate Code to Identify API Usability Problems.*** Daye Nam, Amber Horvath, Andrew Macvean, Brad Myers, and Bogdan Vasilescu, **ASE '19.** [~20% acceptance rate]

[C8] ***Open Source Barriers to Entry, Revisited: A Sociotechnical Perspective.*** Christopher Mendez, Hema Susmita Padala, Zoe Steine-Hanson, Claudia Hilderbrand, Amber Horvath, Charles Hill, Logan Simpson, Nupoor Patil, Anita Sarma, and Margaret Burnett, **ICSE '18.** [21% acceptance rate]

*[C9] **Variolite: Supporting Exploratory Programming by Data Scientists.*** Mary Beth Kery, Amber Horvath, and Brad A. Myers, **CHI '17.** [25% acceptance rate, Best Paper Honorable Mention (top 5%)]

*[C10] **GenderMag Experiences in the Field: The whole, the parts, and the workload.*** Charles Hill, Shannon Ernst, Alannah Oleson, Amber Horvath, and Margaret Burnett, **VL/HCC '16.** [30% acceptance rate]

*[C11] **A Principled Evaluation for a Principled Idea Garden.*** Will Jernigan, *Amber Horvath*, Michael Lee, Margaret Burnett, Taylor Cuilty, Sandeep Kuttal, Anicia Peters, Irwin Kwan, Faezeh Bahmani, and Amy Ko, **VL/HCC '15.** [30% acceptance rate]

*[C12] **To Fix or to Learn? How Production Bias Affects Developers' Information Foraging During Debugging.*** David Piorkowski, Scott D. Fleming, Christopher Scaffidi, Margaret Burnett, Irwin Kwan, Austin Z. Henley, Jamie Macbeth, Charles Hill, and Amber Horvath, **ICSME '15.** [~25.6% acceptance rate]

*[C13] **Principles of a Debugging-First Puzzle Game for Computing Education.*** Michael J. Lee, Faezeh Bahmani, Irwin Kwan, Jilian LaFerte, Polina Charters, Amber Horvath, Fanny Luor, Jill Cao, Catherine Law, Michael Beswetherick, Sheridan Long, Margaret Burnett, and Amy J. Ko, **VL/HCC '14.** [29% acceptance rate]

*[C14] **End-user programmers in trouble: Can the Idea Garden help them to help themselves?*** Jill Cao, Irwin Kwan, Faezeh Bahmani, Margaret Burnett, Scott D. Fleming, Josh Jordahl, Amber Horvath, and Sherry Yang, **VL/HCC '13.** [29.6% acceptance rate]

POSTERS, EXTENDED ABSTRACTS, WORKSHOP PAPERS, AND TECHNICAL REPORTS

*[E1] **Methods for Investigating Mental Models for Learners of APIs.*** Amber Horvath, Mariann Nagy, Finn Voichick, Mary Beth Kery, and Brad A. Myers, **CHI EA '19.** [26% acceptance rate]

Invited Presentations

Program Comprehension in the Era of LLM-Mediated Software Development

- *CSER 2026,* New Faculty Talk Series, Toronto, ON, CA, 2026.
- *University of Victoria*, Victoria, BC, CA, 2026.

Software Comprehension thru Meta-Information: Utilizing User-Derived Data for Developer Support

- *University of Victoria*, Victoria, BC, CA, 2025.

Code as a Material: Utilizing Annotation for Software Comprehension and Development Support with Meta-Information

- IBM HCAI Seminar Series, Boston, MA, 2025.

Meta-Information to Support Sensemaking by Developers

- MIT CSAIL HCI Seminar Series, Boston, MA, 2024.
- University of Illinois-Chicago, Chicago, IL, 2024.
- Microsoft Research, PROSE Team, 2024.
- Depaul University, Chicago, IL, 2024.
- Stevens Institute of Technology, Hoboken, NJ, 2024.

The Long Tail: Understanding the Discoverability of API Functionality

- Stanford HCI Reading Group, Stanford, CA, 2019.

Teaching Experience

6.1040 Software Design, 2025. MIT CSAIL, Honorary Instructor.

05-410/05-610 User-Centered Research and Evaluation, 2022. Carnegie Mellon University HCII, Graduate Teaching Assistant.

05-430/05-630 Programming Usable Interfaces, 2020. Carnegie Mellon University HCII, Graduate Teaching Assistant.

CS 160, 161, and 162 Introduction to Programming series, 2014-2015, 2016-2017. Oregon State University EECS, Undergraduate Teaching Assistant.

Mentorship Experience

Shannon Bonet. REU student, 2022. Graduated from UC Berkeley, BA in Cognitive Science 2023, now Software Engineer at Cadence.

Matthew Shu, REU student, 2022. Graduated from Yale University, BS in Statistics and Sociology 2024, now working on Masters at Yale University.

Connor Shannon, Master's student, 2021. Graduated from Carnegie Mellon University, MA in Human-Computer Interaction 2022, now Front-End Designer at Nava.

Emma Paterson, REU student, 2021. Graduated from Tufts University, BS in Computer Science 2024, now Software Engineer at Microsoft.

Imtiaz Rahman, REU student, 2021. Graduated from Hunter College, BS in Computer Science 2022, now Software Engineer at American Express.

Kazi Jawad, Undergraduate researcher, 2021. Graduated from Carnegie Mellon University, BS in Statistics and Machine Learning 2023, now Technical Director at Weta FX.

Lai Wei, Master's student, 2020. Graduated from Carnegie Mellon University, MA in Human-Computer Interaction 2021, now UX Designer at Amazon.

Service

Faculty EDI Committee, 2026-present. University of Victoria, CS Department representative.

Paper Reviewer, 2020-present. ACM CHI 2022-2027 [5 outstanding reviews]; ACM IUI 2025-2026; ACM UIST 2021, 2023 - 2026 [3 outstanding reviews]; ACM/IEEE ESE 2021; ACM TOCE 2020.

Program Committee Member, 2025-2027. Intelligent User Interface (IUI) Conference.

Speaker, Oregon State University, 2024, 2025. CS capstone course panel on post-graduation career paths.

Application Reviewer and Review Coordinator, 2023-2019. Carnegie Mellon University HCII REU Program.

Application Reviewer, 2022-2018. Carnegie Mellon University S3D REU Program.

Speaker, Apprenticeships in Science and Engineering, 2022. Panel discussion about former intern success stories.

Ph.D Committee Member, 2022. Carnegie Mellon University HCII Director Search.

Ombudsperson, 2021-2022. Carnegie Mellon University HCII Ph.D. program.

Speaker, 2021. Bit-by-Bit conference, an outreach event for high school girls to learn about computer science.

Application Reviewer, 2021. Ph.D. applications by underrepresented people applying to STEM programs.

Student Member, 2019-2020. Software Developer Diversity and Inclusion summit.

Awards and Honors

Diversity and Inclusion Award, 2023. Awarded to our paper "Systemic Gender Inequities in Who Reviews Code" from CSCW '23 [J1].

ARCS Fellowship, 2018-2021. Awarded by the ARCS foundation.

Best Paper Honorable Mention, 2017. Awarded to our paper "Variolite: Supporting Exploratory Programming by Data Scientists" from CHI '17 [C9].

CBT Nuggets Scholarship, 2015-2017. Awarded by CBT Nuggets for excellence in engineering.

Undergraduate Researcher of the Year, Honorable Mention, 2015. Awarded by Oregon State University.

College of Engineering Dean's List Scholarship, 2013-2015. Awarded by Oregon State University.

Aspirations in Computing Award, 2013. Awarded by the National Center for Women in Technology.

References

Brad A. Myers	*Arvind Satyanarayan*	*David Karger*
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Andrew Macvean	*Emerson Murphy-Hill*	*Margaret Burnett*
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Laura Dabbish	*Niki Kittur*	*Elena Glassman*
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